Undetermined coefficients for higher order equations

As we mentioned previously, the solution to a higher order nonhomogeneous equation ($g(t) \neq 0$),

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \ldots + p_1(t)y' + p_0(t)y = g(t)$$

just like for second order equations, will be the sum of the complementary solution $y_c(t)$ (the solution to the associated homogeneous equation) and the particular solution $y_p(t)$.

$$y(t) = y_c(t) + y_p(t)$$

$$y(t) = [c_1y_1(t) + c_2y_2(t) + \ldots + c_ny_n(t)] + y_p(t)$$

So our process for solving a nonhomogeneous equation will be the same process we used for second order equations.

1. Find the complementary solution $y_c(t)$ by solving the associated homogeneous equation.

2. Make a guess for the particular solution $y_p(t)$, eliminating any overlap between the guess and the complementary solution.

3. Take derivatives of the particular solution, then plug the guess and its derivatives into the original differential equation.

4. Solve for the constants A, B, C, etc., then plug their values back into the guess for the particular solution.

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5. Take the sum of the complementary and particular solutions to get the general solution $y(t) = y_c(t) + y_p(t)$ to the nonhomogeneous equation.

And just like before, this method of undetermined coefficients works well when $g(t)$ is some combination of exponential, polynomial, and sine and cosine functions.

Let's do an example so that we can see these steps in action.

Example

Find the general solution to the nonhomogeneous differential equation.

$$y''' - 6y'' + 12y' - 8y = 3e^{2t} + 4e^{3t}$$

We need to by solving the associated homogeneous equation, so we'll set $g(t) = 0$, then factor the characteristic equation associated with the homogeneous equation.

$$r^3 - 6r^2 + 12r - 8 = 0$$

$$(r-2)(r-2)(r-2) = 0$$

We find the repeated root $r = 2$ with multiplicity three, so the complementary solution for the nonhomogeneous equation (or the general solution for the homogeneous equation), is given by

$$y_c(t) = c_1 e^{2t} + c_2 t e^{2t} + c_3 t^2 e^{2t}$$

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Our initial guess for the particular solution should be

$$y_p(t) = Ae^{2t} + Be^{3t}$$

The first term in our guess Ae^{2t} overlaps with the first term from the complementary solution c_1e^{2t}. To eliminate the overlap, we should multiply by t to get Ate^{2t}. However, this new term still overlaps with c_2te^{2t} from the complementary solution, so we'll need to multiply by t again to get At^2e^{2t}, but then by t again to eliminate the overlap with $c_3t^2e^{2t}$ and get At^3e^{2t}.

$$y_p(t) = At^3e^{2t} + Be^{3t}$$

We'll find derivatives of the particular solution,

$$y_p'(t) = 2At^3e^{2t} + 3At^2e^{2t} + 3Be^{3t}$$

$$y_p''(t) = 4At^3e^{2t} + 12At^2e^{2t} + 6Ate^{2t} + 9Be^{3t}$$

$$y_p'''(t) = 8At^3e^{2t} + 36At^2e^{2t} + 36Ate^{2t} + 6Ae^{2t} + 27Be^{3t}$$

and then plug them into the original differential equation.

$$8At^3e^{2t} + 36At^2e^{2t} + 36Ate^{2t} + 6Ae^{2t} + 27Be^{3t}$$

$$-6(4At^3e^{2t} + 12At^2e^{2t} + 6Ate^{2t} + 9Be^{3t})$$

$$+12(2At^3e^{2t} + 3At^2e^{2t} + 3Be^{3t}) - 8(At^3e^{2t} + Be^{3t}) = 3e^{2t} + 4e^{3t}$$

$$6Ae^{2t} + Be^{3t} = 3e^{2t} + 4e^{3t}$$

This gives the system of differential equations

$$6A = 3 \text{ so } A = 1/2$$

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$$B = 4$$

and therefore the particular solution is

$$y_p(t) = \frac{1}{2}t^3e^{2t} + 4e^{3t}$$

and the general solution is the sum of the complementary and particular solutions.

$$y(t) = y_c(t) + y_p(t)$$

$$y(t) = c_1e^{2t} + c_2te^{2t} + c_3t^2e^{2t} + \frac{1}{2}t^3e^{2t} + 4e^{3t}$$

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